Practice Test 3: Solutions & Marking Guide

Total: 20 marks plus 2 bonus. Open this guide only after attempting the paper under test conditions. Follow-through credit applies to a clear method after a minor earlier error.

Question 1 (4)

(a) [2] Use the Laplace transform. The two properties needed are that convolution becomes multiplication and that d/dt becomes multiplication by s (for signals starting at zero). The synapse transforms to

$$H(s) = \int_0^\infty \tfrac{1}{\tau} e^{-t/\tau} e^{-st} dt = \frac{1}{1+s\tau}.$$

Then

$$X = \frac{G+F}{1+s\tau} \;\Rightarrow\; X + s\tau X = G + F \;\Rightarrow\; s\tau X = G + F - X \;\Rightarrow\; \boxed{\dot{x} = \tfrac{1}{\tau}(g(u) + f(x) - x)}$$

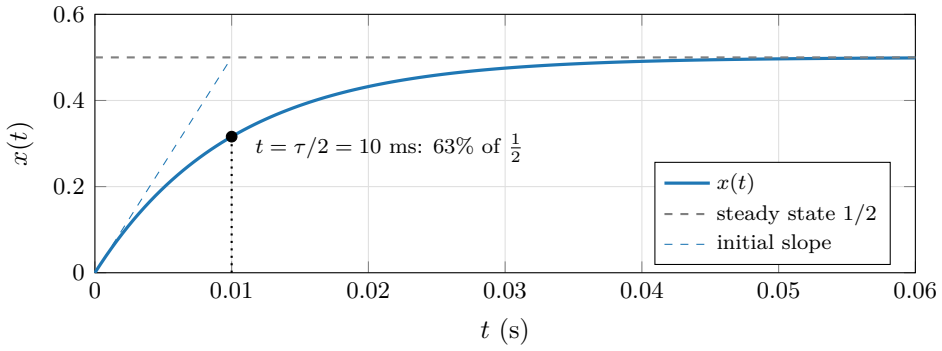
Marking

0.5: names the Laplace transform and both properties. 0.5: $H(s) = 1/(1+s\tau)$ (evaluated or correctly quoted). 1: algebra to the final ODE, including the $-x$ term. A correct time-domain derivation (differentiating the convolution integral) earns full marks.

(b) [2] $\tau\dot{x} = u - x - x = 1 - 2x$, so $\dot{x} = \frac{1}{\tau/2}(\frac{1}{2} - x)$ and

$$x(t) = \tfrac{1}{2}(1 - e^{-t/(10\,\text{ms})}).$$

The steady state is $\frac{1}{2}$ and the time constant is $\tau/2 = 10$ ms, twice as fast as the synapse alone.



Marking

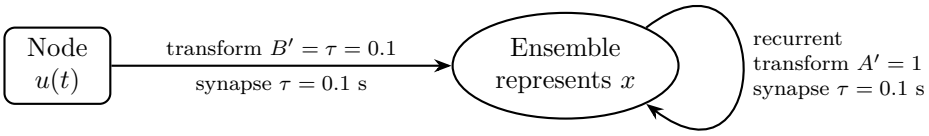
0.5: starts at 0 and rises concavely (steepest at $t = 0$), no overshoot. 0.5: asymptote $\frac{1}{2}$ labelled. 0.5: time constant 10 ms labelled (as $\tau/2$ or 63% point). 0.5: axes labelled with units. A steady state of 1 or time constant 20 ms loses the corresponding half-mark.

Where marks are usually lost

The steady-state guess $x = u - x$ gives $\frac{1}{2}$ but not the speed. The time constant is not the synapse's 20 ms: the feedback $-x$ doubles the leak.

Question 2 (4)

(a) [2] $A = 0$, $B = 1$. With $A' = \tau A + I$ and $B' = \tau B$: $A' = 1$ and $B' = \tau = 0.1$.



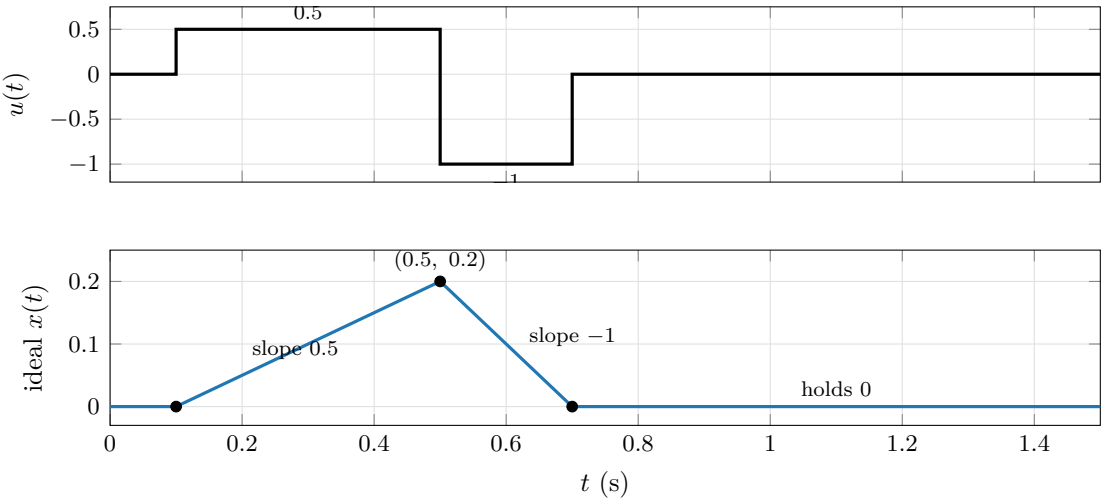
Marking

0.5: $A = 0$, $B = 1$. 0.5: $A' = 1$, $B' = 0.1$. 1: diagram with an input connection and a recurrent connection back onto the same ensemble, each labelled with transform and synapse. Swapping A' and B' loses the second half-mark.

(b) [2]

$$x(t) = \int_0^t u(t')\,dt' = \begin{cases} 0, & t < 0.1, \\ 0.5(t - 0.1), & 0.1 \le t < 0.5, \\ 0.2 - (t - 0.5), & 0.5 \le t < 0.7, \\ 0, & t \ge 0.7. \end{cases}$$

Check the joins: at $t = 0.5$, $0.5(0.4) = 0.2$; at $t = 0.7$, $0.2 - 0.2 = 0$.



Marking

1: all four cases, each starting from the previous end value, with no integration constant (0.5 for the middle two cases, 0.5 for the outer two). 1: aligned sketch with correct slopes, peak 0.2 at $t = 0.5$, return to 0 at $t = 0.7$, flat afterward, axes labelled.

Where marks are usually lost

Writing $x = -t$ on the third piece (forgetting it starts from 0.2), and drawing x dropping back to zero when u turns off. An integrator holds its value when $u = 0$.

Question 3 (4)

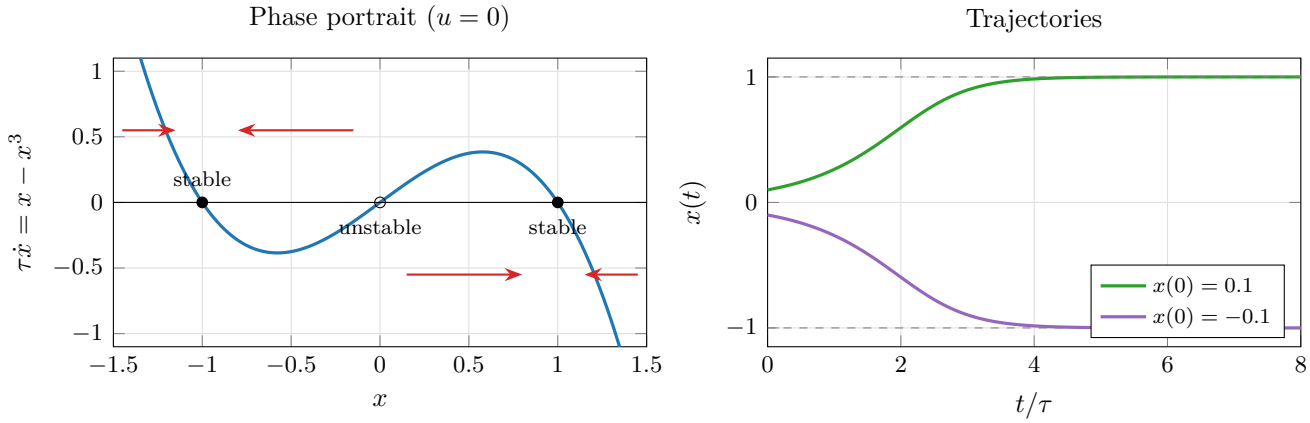
(a) [2] $\tau\dot{x} = u + f(x) - x = 2x - x^3 - x = x - x^3 = x(1-x)(1+x)$. Equilibria: $x = 0, \pm 1$. With $\tau\phi'(x) = 1 - 3x^2$: at $x = 0$ it is $+1$, so 0 is unstable. At $x = \pm 1$ it is -2, so ± 1 are stable.

Marking

0.5: correct $\dot{x} = (x - x^3)/\tau$, including the $-x$ leak. 0.5: all three equilibria. 1: correct stability with a reason (sign of ϕ', or arrows on each side). Classifying with the algebraic $x = f(x)$ alone earns no stability marks.

(b) [2] Between 0 and 1, $\dot{x} > 0$ (moves right); beyond 1, $\dot{x} < 0$ (moves left). The picture is mirrored for negative x. With $u = 0$ the state is pushed away from 0 to whichever of ± 1 is on its side:

$$x(t) = \frac{x_0 e^{t/\tau}}{\sqrt{1 - x_0^2 + x_0^2 e^{2t/\tau}}}.$$



It is a memory because with no input the state stays at $+1$ or -1 indefinitely, so the network stores which way it was last pushed (one bit).

Marking

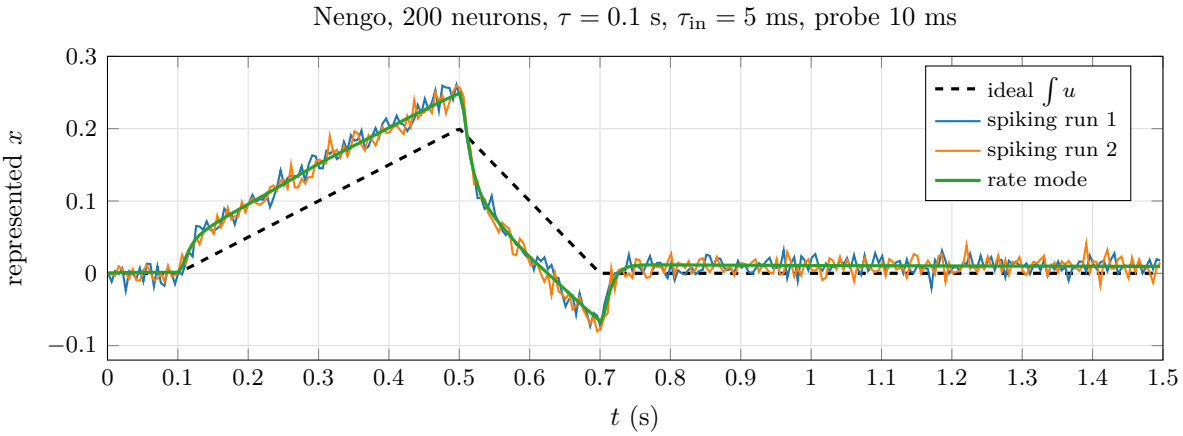
1: cubic shape with three zeros and correctly directed arrows, stable and unstable points marked, axes labelled. 0.5: both trajectories leave ± 0.1 slowly at first, then turn over and settle at ± 1 (S-shaped, no crossing of zero). 0.5: memory sentence (persistence without input, history-dependent). Exact timing is not required.

Question 4 (4)

(a) [2]

- **Static distortion (decoder error).** The recurrent decoder gives $\hat{f}'(x) \neq x$, so the real dynamics are $\dot{x} = (\hat{f}'(x) - x)/\tau \neq 0$. This has small stable and unstable zeros, and the state drifts deterministically toward the nearest stable one. That is the repeatable rate-mode drift.
- **Spike noise.** Spikes make the filtered activity fluctuate, so the state takes a random walk. That is why spiking runs wander and end at different values.

More neurons reduce both. Static distortion shrinks roughly as $1/n$ (in RMSE), so the drift slows. Spike noise averages out roughly as $1/\sqrt{n}$, so the random walk shrinks. Neither reaches zero.



After $t = 0.7$ s the spiking traces wander around differently, while the rate trace drifts smoothly and slowly. The offset during the input is part (b).

Marking

0.5 + 0.5: each source named. 0.5: correct mapping (distortion gives the deterministic drift that survives rate mode; spike noise gives the run-to-run variation). 0.5: both shrink with more neurons. Saying "noise" for both sources earns at most 1.

(b) [2] With $A' = 1$ and $B' = \tau$:

$$X = \frac{X}{1+s\tau} + \frac{\tau U}{1+s\tau_{\text{in}}} \;\Rightarrow\; X\frac{s\tau}{1+s\tau} = \frac{\tau U}{1+s\tau_{\text{in}}} \;\Rightarrow\; X = \frac{1+s\tau}{1+s\tau_{\text{in}}} \cdot \frac{U}{s}.$$

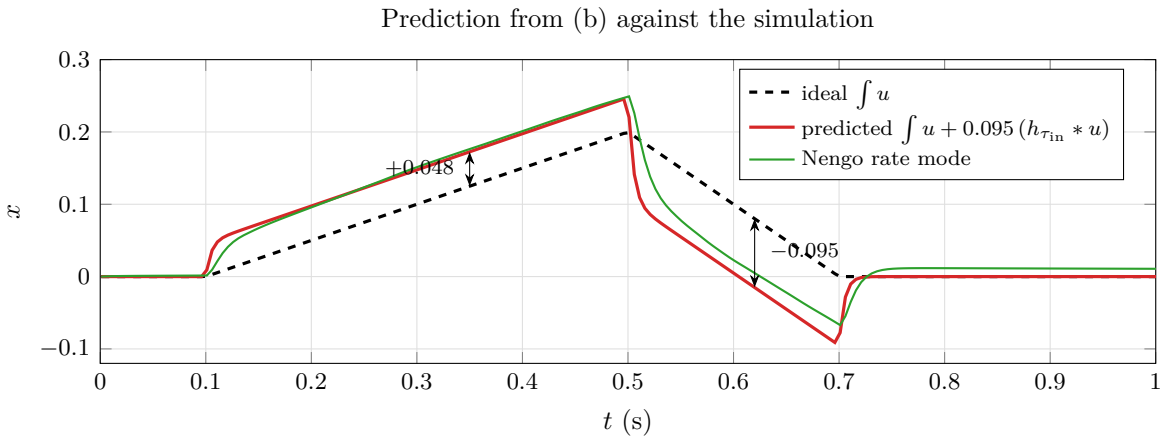
Since $\frac{1+s\tau}{1+s\tau_{\text{in}}} = 1 + \frac{s(\tau-\tau_{\text{in}})}{1+s\tau_{\text{in}}}$,

$$X = \frac{U}{s} + (\tau - \tau_{\text{in}})\frac{U}{1+s\tau_{\text{in}}}, \qquad x(t) = \int_0^t u + (\tau - \tau_{\text{in}})\,(h_{\tau_{\text{in}}} * u)(t).$$

The integral is exact. On top of it, the input leaks into the state, filtered by 5 ms and weighted by $\tau - \tau_{\text{in}} = 0.095$ s:

- $0.1 \le t < 0.5$: about $+0.095 \times 0.5 \approx +0.048$ above the ideal (so about 0.25 instead of 0.2 at $t = 0.5$);
- $0.5 \le t < 0.7$: about -0.095 below the ideal;
- after 0.7: the offset decays within a few ms (several τ_{in}) and the final value is the ideal 0.

Each jump of u produces a quick step of $0.095\,\Delta u$ in the trace, smoothed over about 5 ms.



The simulated trace follows the prediction, plus the smoothing of the 10 ms probe filter.

Marking

1: correct Laplace setup with two different filters and algebra to the stated form. 0.5: offset $(\tau - \tau_{\text{in}})u$ with sizes $\approx +0.048$ and -0.095. 0.5: offset disappears after the input ends; the final value is unaffected.

Where marks are usually lost

Claiming the only effect is "a small 5 ms lag". The leak-through term is 19 times larger than τ_{in} would suggest, because it is weighted by $\tau - \tau_{\text{in}}$.

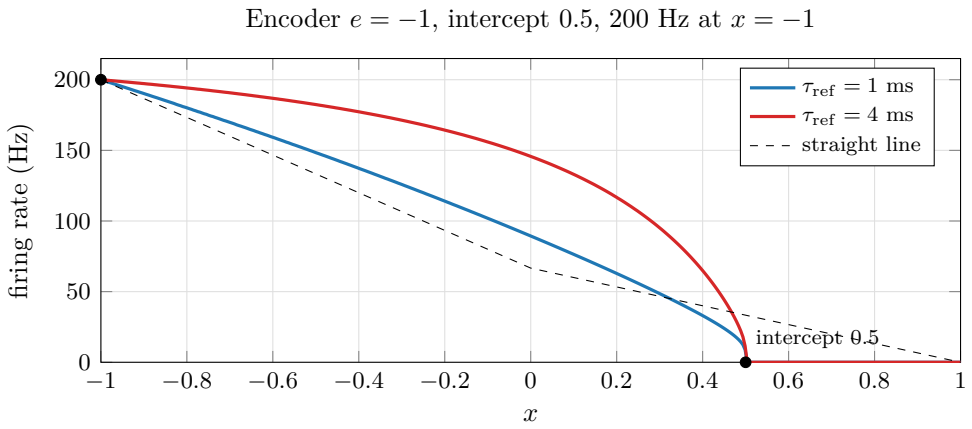
Question 5 (4)

(a) [1] RMSE $= 3\varepsilon$. Nengo represents x/r internally with the same neurons, so the normalized decoding problem is identical and the decoded output, including its error, is multiplied by r.

Marking

0.5: 3ε (or RMSE $\propto r$). 0.5: the scaling reason.

(b) [2] With intercept and maximum rate fixed, $J(x) = 1 + (J^{\max} - 1)\frac{0.5-x}{1.5}$ for $x < 0.5$, where $J^{\max} = \left(1 - e^{(\tau_{\text{ref}} - 1/200)/\tau_{RC}}\right)^{-1}$: ≈ 5.5 for $\tau_{\text{ref}} = 1$ ms and ≈ 20.5 for 4 ms. Both curves are 0 for $x \geq 0.5$ and reach 200 Hz at $x = -1$. The 4 ms curve shoots up just left of the intercept and then flattens, because at 200 Hz it is close to its $1/\tau_{\text{ref}} = 250$ Hz ceiling. The 1 ms curve (ceiling 1000 Hz) is much straighter.



The population with the larger τ_{ref} has the larger RMSE. Its more curved, saturating tuning curves are worse building blocks for the straight line $\hat{x} = x$. Nengo, for 100 neurons: about 0.0053 at 1 ms against 0.0069 at 4 ms.

Marking

0.5: both curves decrease toward the right, zero for $x \geq 0.5$ (correct encoder direction). 0.5: same intercept and same 200 Hz endpoint at $x = -1$; 4 ms curve visibly more concave. 1: larger τ_{ref} gives larger RMSE, with a reason linking curvature or saturation to decoding error. Drawing ReLU-style straight lines loses the shape half-mark.

(c) [1]

$$v = (0.2, 0.5) - 2(0.3, -0.1) + 3(0.1, 0.2) = (0.2 - 0.6 + 0.3,\ 0.5 + 0.2 + 0.6) = (-0.1,\ 1.3),$$

so $\|v\| = \sqrt{1.70} \approx 1.30$. This lies outside the radius-1 disc, so $\hat{v}$ will be inaccurate: the second component falls short and the error is far larger than inside the trained range. Fix: give v a radius of at least about 1.3 (for example 1.5), or scale the transforms down and rescale downstream.

Marking

0.5: $v = (-0.1, 1.3)$. 0.5: notes $\|v\| > 1$ and gives a valid fix. Checking only the components against 1 is acceptable here, since $1.3 > 1$ either way.

Bonus Question 1 (2)

The synapse is a leaky integrator. It drifts toward whatever input it receives and forgets its own value with time constant τ. The "$+I$" feeds the current state back into the population, which exactly replaces what the synapse leaks away, so with nothing else added the state holds. Multiplying by τ converts the desired rate of change into the input the slow synapse needs, because a synapse moves by (input − state)$/\tau$ per second. A feed-forward chain has no such loop. Each population's state is just its filtered input, so it decays within a few synaptic time constants (tens of milliseconds) after the input stops.

Marking

0.5: synapse as leaky integrator / forgetting. 0.5: $+I$ cancels the leak. 0.5: τ scaling as converting a rate into the synapse's input. 0.5: feed-forward forgets within a few τ.